

DL Graded Quiz M4

Question 1

Suppose you are working with an RNN model without bias terms. The first RNN layer of this model has 25 neurons. The second (subsequent) RNN layer has 25 neurons as well. You are planning to use a window for a sequence of 36 in the 3 input feature of the model. How many parameters will your model need to train in the first RNN layer?

- ☐ 900
- ☐ 700
- ☐ 525
- ☒ 725

Question 2

Which of the following statements is **true**?

- ☐ The parameterization cost of an RNN grows as the number of time steps increases.
- ☐ The parameterization cost of an RNN is reduced as the number of time steps increases.
- ☐ The parameterization cost of an RNN does not grow as the number of time steps increase.
- ☒ The parameterization cost of an RNN grows as the number of time steps decreases.

Question 3

Under a latent variable model, how would we model the following probability?

$$P(\mathbf{x}_t | \mathbf{x}_{t-1}, \dots, \mathbf{x}_1)$$

- ☐ $P(\mathbf{x}_t | \mathbf{h}_{t-1})$
- ☐ $P(\mathbf{x}_t | \mathbf{h}_t)$
- ☐ $P(\mathbf{x}_t)P(\mathbf{x}_{t-1})P(\mathbf{x}_{t-2}) \dots P(\mathbf{x}_1)$
- ☒ $P(\mathbf{x}_t)P(\mathbf{x}_{t-1} | \mathbf{h}_{t-1})$

Question 4

How could you also express the following gradient?

$$\frac{\partial L}{\partial W_{xh}}$$

- ☐ $\sum_{t=1}^T \text{prod} \left(\frac{\partial L}{\partial W_{xh}}, \frac{\partial W_{xh}}{\partial o_t} \right)$
- ☐ $\sum_{t=1}^T \text{prod} \left(\frac{\partial L}{\partial h_t}, \frac{\partial h_t}{\partial W_{xh}} \right)$
- ☐ $\sum_{t=1}^T \text{prod} \left(\frac{\partial L}{\partial W_{xh}}, \frac{\partial h_t}{\partial o_t} \right)$
- ☒ $\text{prod} \left(\frac{\partial L}{\partial h_t}, \frac{\partial h_t}{\partial W_{xh}} \right)$

Question 5

Suppose you are working with a language model that use sentences as inputs. Suppose the input sentence is "Hello WorldQuant friends!", on which you plan to use regular truncation by setting $\tau = 5$. How many subsequences would you obtain?

- ☐ 1
- ☐ 4
- ☐ 5
- ☒ 3

Question 6

How could you represent a one-hot vector with indices 1 and 3 in 4-element vocabulary? (Assume that all answers are in the form of tensors.)

- ☐ $[1, 0, 0, 0][0, 0, 1, 0]$
- ☐ $[1, 0, 3, 0][0, 1, 0, 1]$
- ☐ $[0, 1, 0, 0][0, 0, 0, 1]$
- ☒ $[0, 1, 0, 3][0, 1, 0, 1]$

Question 7

Under a first-order Markov condition, what does the following formula represent?

$$P(x_1) \prod_{t=2}^T P(x_t | x_{t-1})$$

- ☐ Joint probability density of sequence $(x_1, \dots, x_T)$
- ☐ Sum of conditionally independent probabilities
- ☐ Probability of x_1 in the sequence
- ☒ Probability of x_2 in the sequence

Question 8

In the following expression, what do large powers of W_{hh}^T cause?

$$\sum_{i=t}^T (W_{hh}^T)^{T-i} W_{gh}^T \frac{\partial L}{\partial o_{T+t-i}}$$

- ☐ Higher learning rate
- ☐ Numerical instability
- ☐ Higher speed of convergence
- ☒ Slower speed of convergence

Question 9

The VIX index is constructed using which of the following?

- ☐ ETF prices from SPY
- ☐ Prices of futures on the S&P500
- ☐ Historical volatility from options on the S&P500 index
- ☒ Implied volatility from options on the S&P500 index

Question 10

How could you represent a one-hot vector with indices 0, 3, and 6 in 8-element vocabulary? (Assume that all answers are in the form of tensors.)

- ☐ $[1, 0, 0, 0, 0, 0, 0, 0][0, 1, 0, 0, 0, 0, 0, 0][0, 0, 1, 0, 0, 0, 0, 0]$
- ☐ $[1, 0, 0, 0, 0, 0, 0, 0][0, 0, 0, 1, 0, 0, 0, 0][0, 0, 0, 0, 0, 0, 1, 0]$
- ☐ $[1, 0, 0, 1, 0, 0, 1, 0]$
- ☒ $[1, 0, 0, 0, 0, 0, 0, 0][0, 0, 0, 3, 0, 0, 0, 0][0, 0, 0, 0, 0, 0, 6, 0]$

Question 11

Suppose a hidden layer in an RNN of the usual form:

$$H = \phi(XW_{xh} + H_{t-1}W_{hh} + b_h)$$

X is a 5x1 matrix, W_{xh} is 1x10, and H_{t-1} is 5x4. What should be the size of W_{hh} ?

- ☐ 10 x 5
- ☐ 4 x 10
- ☐ 4 x 5
- ☒ 5 x 4

Question 12

When calculating the following gradient:

$$\frac{\partial L}{\partial h_t} = \sum_{i=t}^T (W_{hh}^T)^{T-i} W_{qh}^T \frac{\partial L}{\partial o_{T+t-i}}$$

where does the problem of long sequences especially impact the computation?

- ☐ Large sequence will mean large powers of W_{hh}^T
- ☐ Large sequence will mean tiny powers of W_{qh}^T
- ☐ Large sequence will mean large powers of W_{qh}^T
- ☒ Large sequence will mean tiny powers of W_{hh}^T

Question 13

Suppose you are using an RNN and dealing with the fourth hidden state, h_4 . Under the usual notation, which of the following variables will help determine h_4 ?

- ☐ W_{hx}, W_{xh}, x_3, h_4
- ☐ $W_{xh}, W_{hh}, x_4, h_3, o_3$
- ☐ W_{hx}, W_{hh}, x_5, h_4
- ☒ W_{hx}, W_{hh}, x_4, h_3

Question 14

What is a positive (desirable) side effect of using the following gradient clipping heuristic?

$$g \leftarrow \min \left(1, \frac{\theta}{\|g\|} \right) g$$

- ☐ Convergence is always achieved



Time left:
2:59:37

QUESTIONS

- | | | | |
|----|----|----|----|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

- ☐ Learning rates decrease with training
- ☐ Each minibatch has a large effect on the parameter vector
- ☒ Limiting the influence of any given minibatch

Question 15

In a latent autoregressive model, we can estimate

$$\hat{x}_t = P(x_t|h_t)$$

What does h_t represent?

- ☐ History of past observations
- ☐ Number of sequences needed to predict x_t
- ☐ Expectation on future values of x_t
- ☒ The length of the sequence

Question 16

In practice, randomized truncation does not really work much better than regular truncation. A reason for this is which of the following?

- ☐ The reduced variance of randomized truncation methods harms the accuracy in the computation of gradients.
- ☐ Randomized truncation has a regularizing effect.
- ☐ Increased variance counteracts the accuracy of computing the gradient.
- ☒ Regular truncation computes a full sum whereas randomized truncation only approximates it.

SUBMIT